EART 11300 Environmental Science Practical

Identification of benthic macroinvertebrates- Part 2

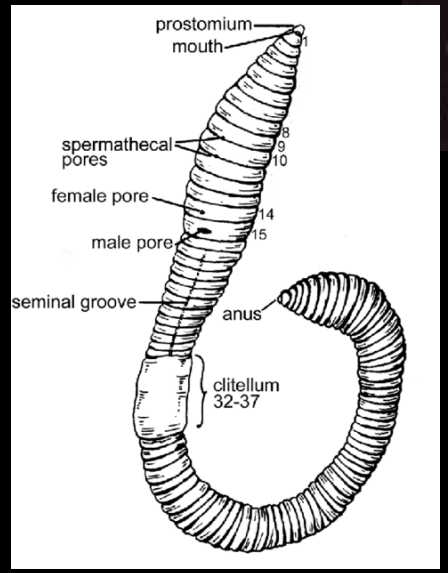


Dr Cecilia Medupin

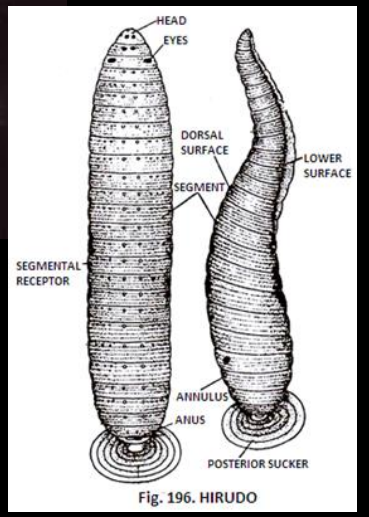
River Aire at Gordale Beck,Yorkshire Dales

Segmented worms (Phylum Annelida)









Subclass Oligochaeta:

clear segmentation

Subclass Hirudinea:

suckers at each end

Leeches

- Narrower towards the front
- Small **sucker** at the end, wider towards the back
- At the front of the **dorsal** side are the eyes- visible as dark spots
- Moves by stretching out the body
- Pollution Score = 3

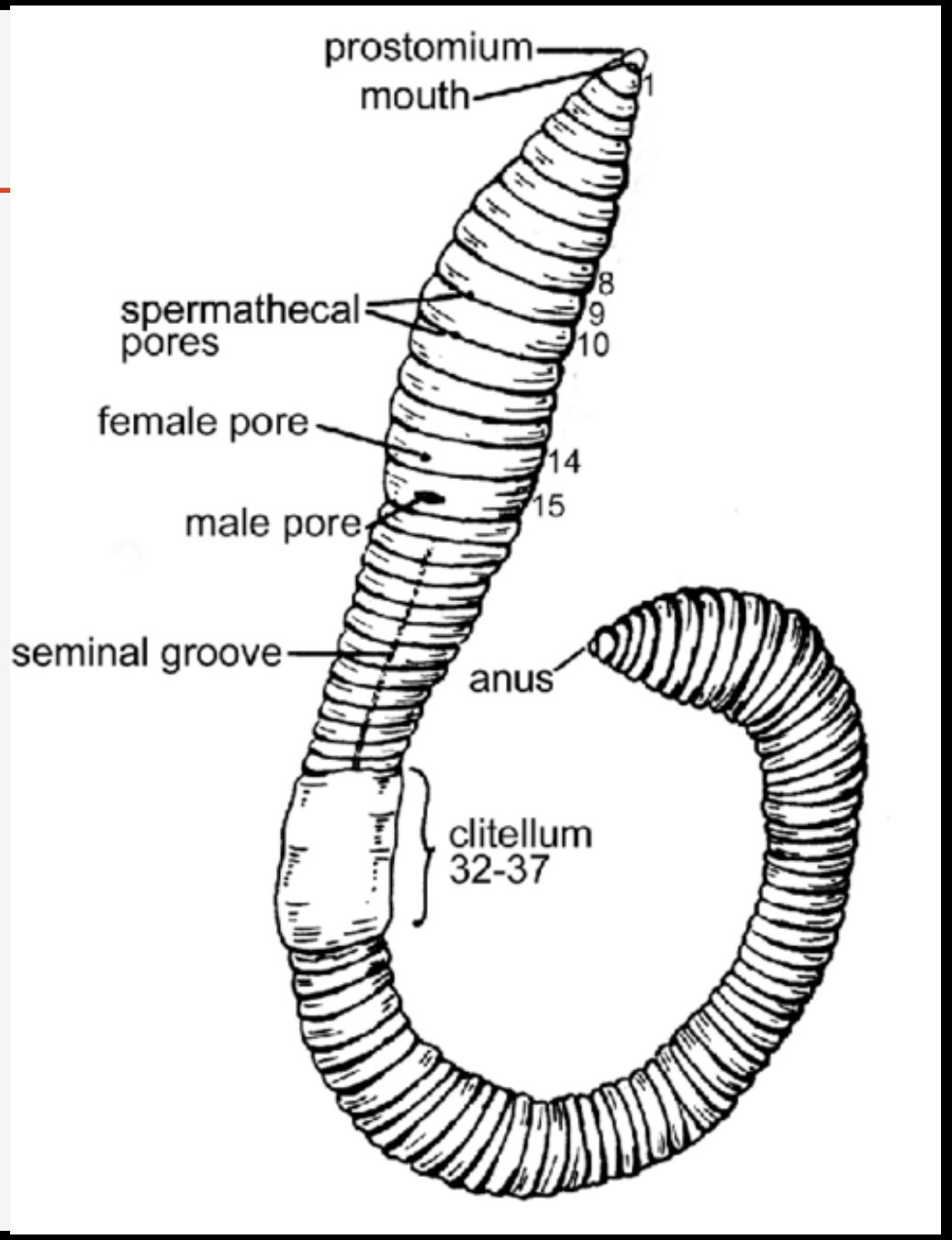




By Prof. Dr. Ulrich Kutschera - Prof. Dr. Ulrich Kutschera - Archivbild, CC BY-SA 3.0, https://commons.wikimedia.org/w/index.php?curid=27729540

Oligochaeta (Worms)

- Has multiple segments
- Based heavily on the **chaetae**
  - the chaetae are stiff bristles, hair-like fibers, made of chitin
  - These hair-like or hook-like features are found on most of the body segments
- Chaetae (could be single or multiple pointed) require high magnification microscope to see them
- Anterior end is thicker than the posterior end
- The very front is the location of a rounded (or sometimes proboscis-like) lobe called the prostomium
- The **clitellum** is a swollen area in which sexual organs are situated . The clitellum is always situated towards the front of the animal if present
- The segments are numbered from front to rear
- Pollution Score = 1

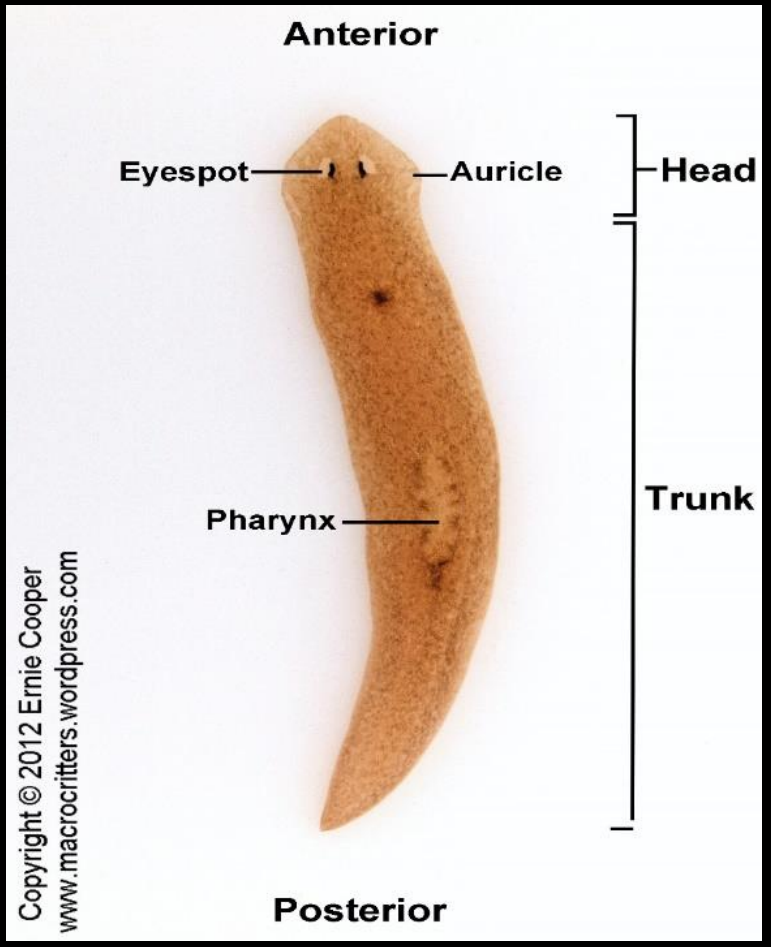


Phylum Platyhelminthes
Flatworms (Tricladida)

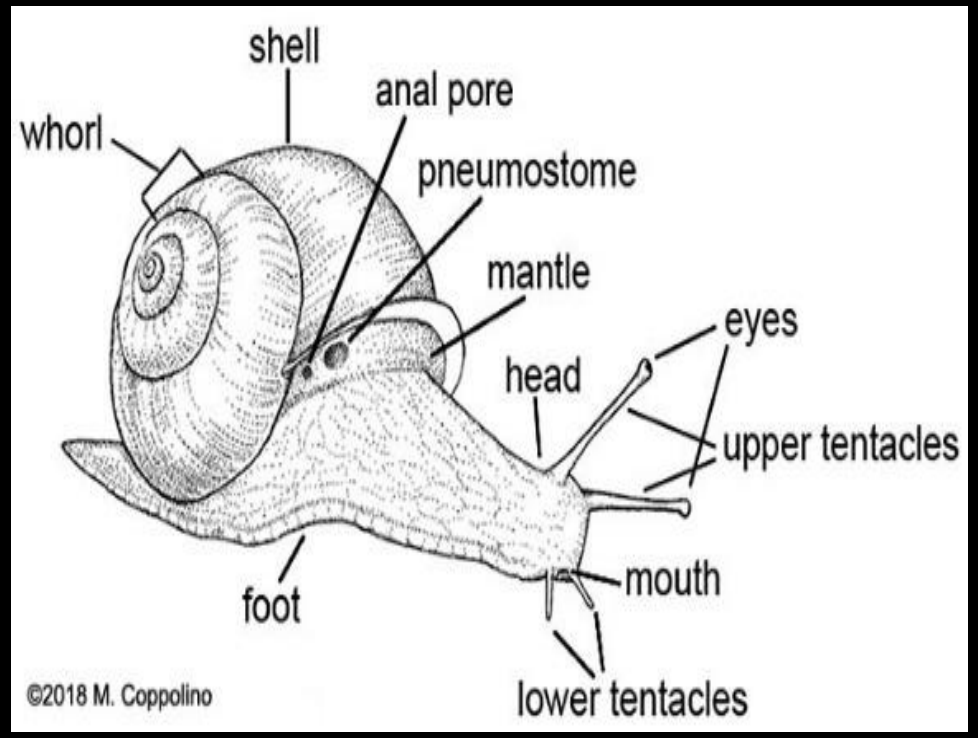
Un-segmented: No obvious head but eyes present;
Distinct gliding movement

- Has multiple segments
- Often black or grey
- Flat because they lack a fluid – filled body cavity
- Bilaterally symmetrical
- Soft bodied
- Moves by gliding
- Pollution Score = 5





Molluscs (Phylum Mollusca) snails & limpets (Gastropoda)











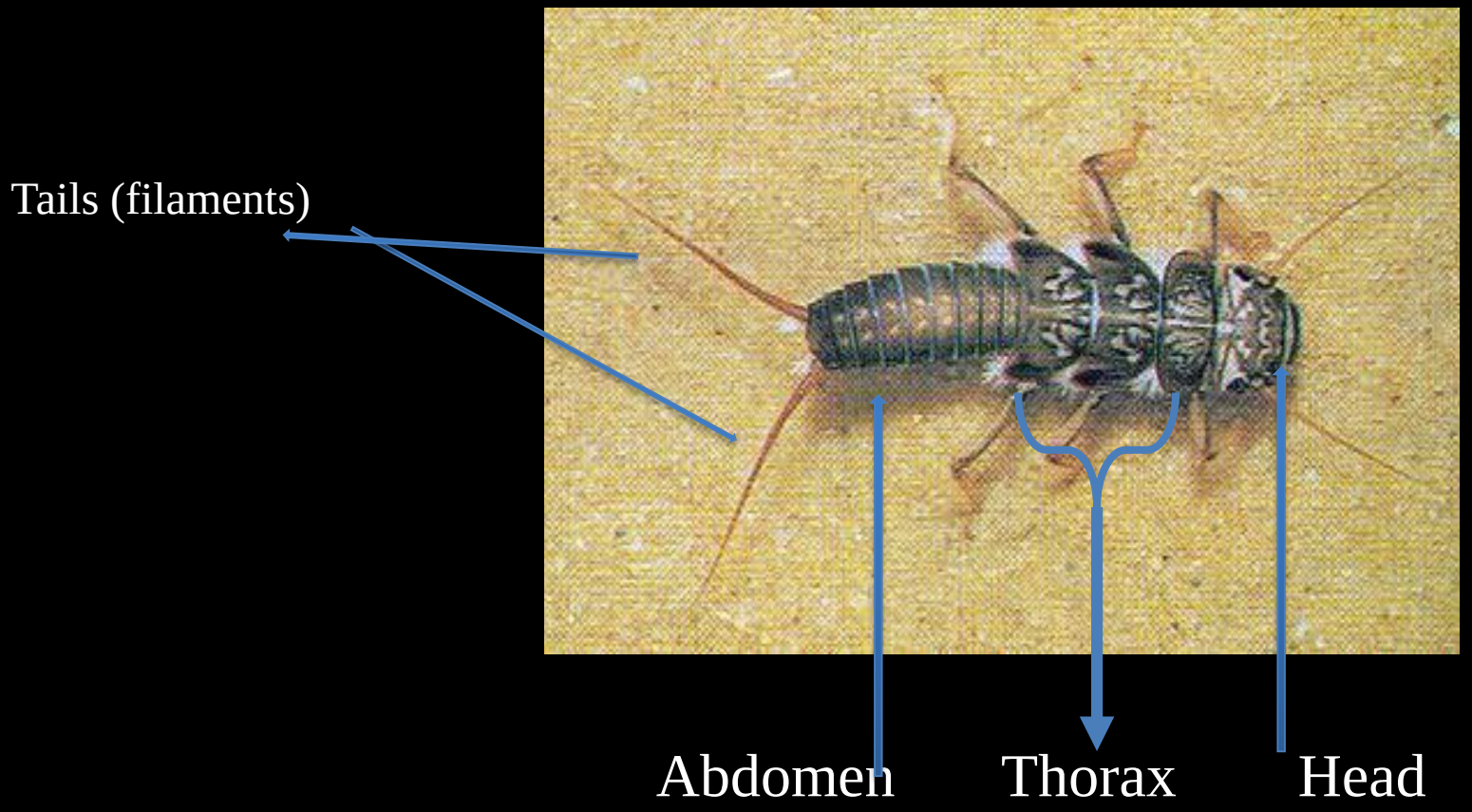
- Hard shell covering body
- Shell may be pointed, flat or coiled
- Moves with the aid of stomach
- Pollution score =3

ID features: shell and aperture shape.

Phylum Arthropoda

From the Greek words “arthros” – joint; “pous” – foot

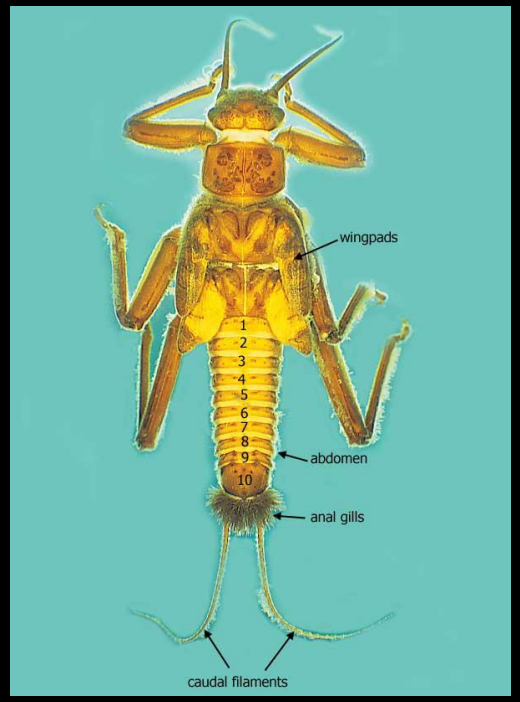
Clear division of segments into different body parts



- The Phylum has the highest number of river organisms
- Has an exoskeleton (chitin)
- Segmented body

Class Insecta

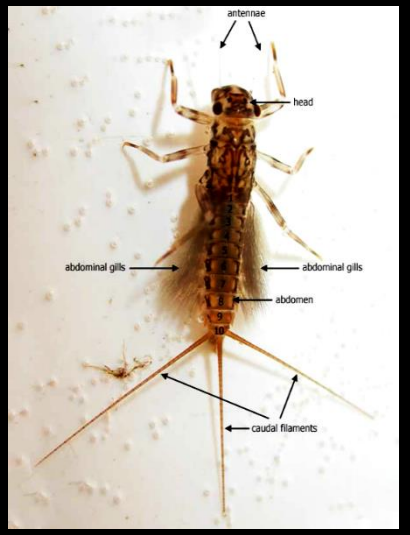




- Stonefly **(Order Plecoptera);** Greek: “pleco” meaning folded
- Have variety of nymphs
- No gills on abdominal segments

“ptera” means wing



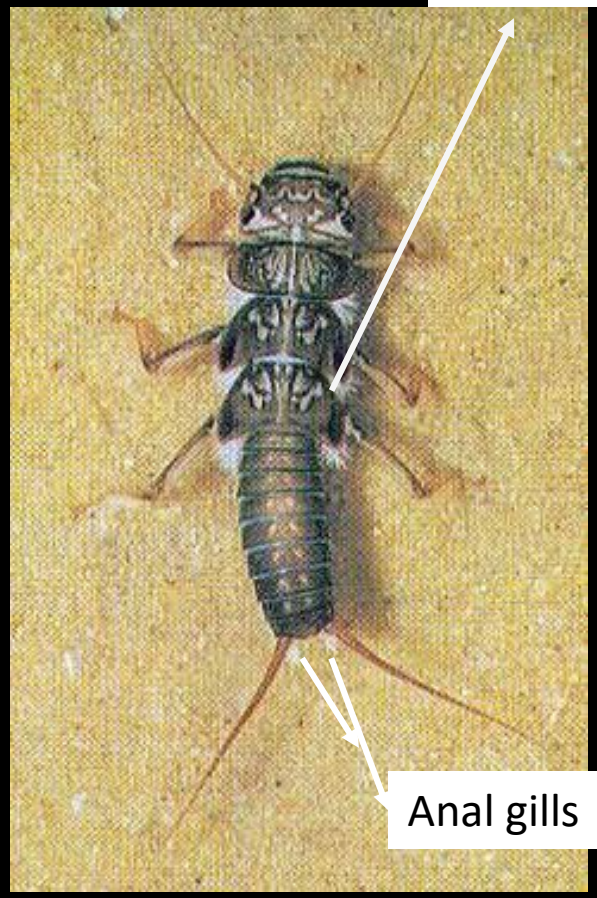


Three filaments
Mayfly (**Order Ephemeroptera**)
Greek: “ephemeros” meaning short lived

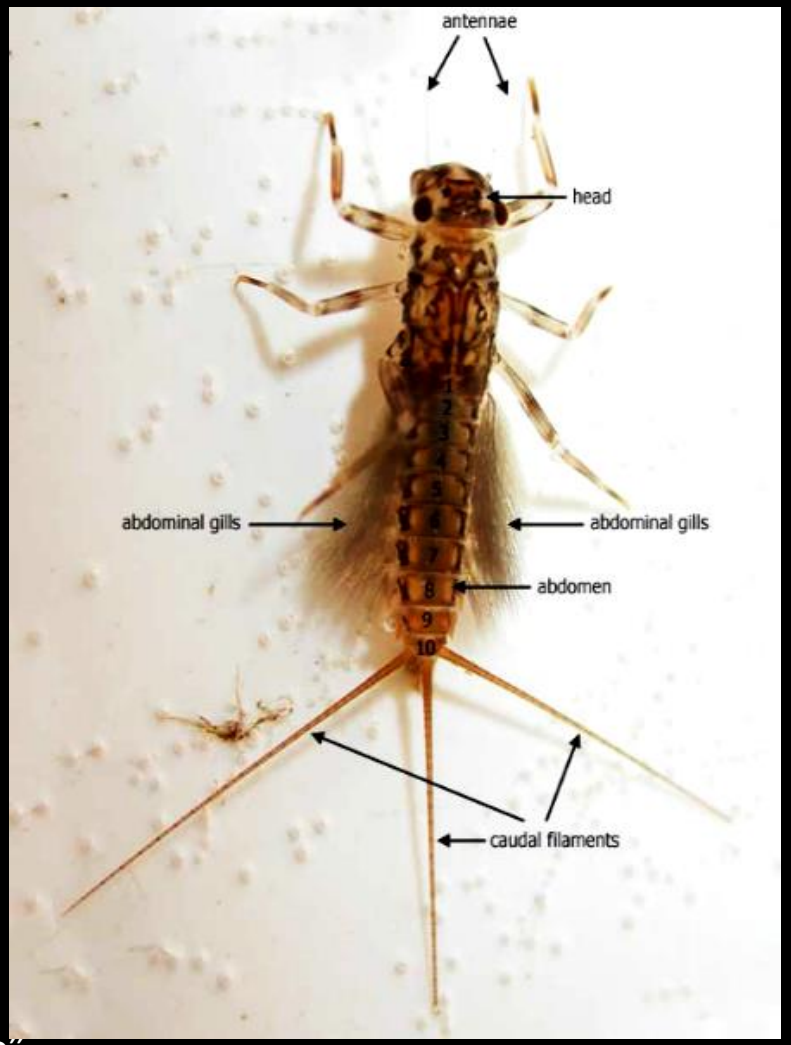


Caddisfly **(Order Trichoptera)**
Latin: “tricho” meaning hairy

Class Insecta



- Stonefly **(Order Plecoptera);** Greek: “pleco” meaning folded; “ptera” means wing
- Have variety of nymphs
- No gills on abdominal segments
- Generally moves by crawling
- Pollution score =10



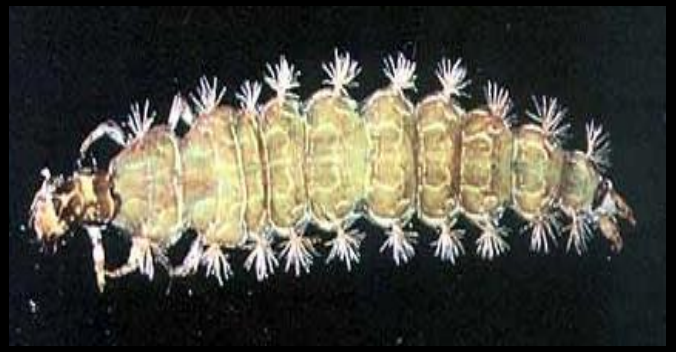
- Mayfly (**Order Ephemeroptera**)
- Greek: “ephemeros” meaning short lived
- Three filaments
- Swims in short darting movements
- Pollution score =10



Caddisfly **(Order Trichoptera)**
Latin: “tricho” meaning hairy

- Body is within a case
- Insects hide in the case undisturbed
- Cases can be made of stones, sand grains, or plant material
- Look for sticks that crawl
- Pollution score =7

Caseless caddisflies (Order Trichoptera)



Rhyacophilidae, Freeliving caddisflies



Hydropsychidae, Common Netspinner caddsflies

- Body is out of a case
- Looks like a thin caterpillar
- Has 3 pairs of legs in the front
- Hooks at the end of body
- Crawls rather then swim
- Pollution score =7



Sedgeflies

Class Insecta
Order Diptera

Diptera – fixed head capsule



Diptera: true flies

- Has considerably fewer segments than Oligochaeta
- Head has no eyes
- Can be red, green, brown or transparent
- It could be red when alive
- Head separated from the thorax
- Found in aquatic, decaying organic substrate
- Pierces and sucks
- Has a wriggling movement
- Pollution Score = 3

Di- 2, “ptera” = wings (named based on the number of wings.



Blackfly larva (Simulidae)



Prolegs

Non-biting midge larva (Chironomidae)

Crustacea

- Characterized by 2 antennae
- Some are vestigial –i.e. hidden by an external carapace
- Some antennae are adapted as feeding or even locomotory organs so they are difficult to identify as antennae
- The antennae is a sensory organ found in most arthropods that occur in pairs attached to the anterior body segment in the head.
- Antennae are normally sub-divided into a series of smaller units (segments)
- Crustaceans have 2 pairs of antennae, insects have a pair



Water louse (Isopoda)



Crayfish (Decapoda)



Water shrimp (Amphipoda)

- Curved, body is flattened
- May appear slightly transparent
- Swims quickly on their sides
- Pollution score =6

Crustacea e.g. water louse

- Characterized by 2 antennae
- Some are vestigial –i.e. hidden by an external carapace
- Some antennae are adapted as feeding or even locomotory organs so they are difficult to identify as antennae
- The antennae is a sensory organ found in most arthropods that occur in pairs attached to the anterior body segment in the head.
- Antennae are normally sub-divided into a series of smaller units (segments)
- Crustaceans have 2 pairs of antennae, insects have a pair
- Body is flattened
- Moves by crawling
- Pollution score = 3



Water louse (Isopoda)

Practice Questions

1. Identify key parts of the following organisms
    a. water louse
    b. Snail
    c. Leech
    d. Insect

2. What are the major morphological differences between
    a. a water louse and a freshwater shrimp?
    b. A worm and a leech?

Assessing the health of rivers – Basis of class workshop

BMWP		ASPT	
BMWP Score	**Biological Quality**	**ASPT**	**Water quality**
Over 130	A. Very good biological quality (natural)	Over 7	Very good (natural)
81 – 130	B. Good biological quality	6.0 – 6.9	Good
51 – 80	C. Fair biological quality	5.0 – 5.9	Fair
11 – 50	D. Poor biological quality	4.0 – 4.9	Poor
0 – 10	E. Very poor biological quality	3.9 or less	Very poor

Biological Monitoring Working Party Score (BMWP) and Average Score Per Taxon

Workshop exercise- Refer to worksheet

- The following benthic macroinvertebrates were identified and counted from two local UK rivers in London and at Leeds: **London**: 10 leeches, 15 snails, 2 stoneflies, 35 freshwater shrimps, 3 flatworms, 25 caseless caddisflies and 2 damselflies; **Leeds:** 22 cased caddisfly larvae, 35 mayflies, 10 leeches, 4 snails, 7 stoneflies, 17 freshwater shrimps and 3 black fly larvae [Note that different types of organisms for individual taxa have not been included in this question].

Questions

1. What are the total BMWP scores and ASPT for both locations? (ASPT score (i.e. total BMWP score divide by the number of taxa)

2. Are there differences between your answers for both locations?

3. From your score, what is the health of the river: very poor, poor, fair, good, or very good? …… (refer to table 1)

Summary

1. Taxonomy- study of naming, identifying & classifying groups of biological organisms based on shared characteristics
2. Benthic macroorganisms that inhabit bottom substrates for at least part of their life cycle and are retained by a 200µm to 500µm mesh.
3. Identified parts of a dissecting microscope e.g. lens, eye piece, stage etc
4. Identified some benthic macroinvertebrates and can differentiate between some of them
5. Briefly learnt that benthic macroinvertebrates can be used to assess river health

The End, Questions

